STUDY OF THE RIEMANN HYPOTHESIS

James Rosado

Complex Analysis: Professor Thomas Osler

Spring 2014

The Riemann Hypothesis states that the non-trivial zeros of the ζ-function lie along a critical line in the complex plane with the real part equal to ½; in other words, the non-trivial zeros are of the form $z = ½ + iy$ [Edwards: Riemann, p301]. As a result of this statement, there have been many equivalent versions of the hypothesis and extensions involving prime numbers and prime number distributions. In this paper I will address some of the various hypotheses that have been written and provide detailed steps on how these hypotheses were arrived.

1. "$\pi(x)$ AND THE ZEROS OF $\zeta(z)$"

From Amer Iqbal's paper "Prime Numbers and Riemann Zeta Function," Iqbal establishes a relationship between the $\pi(x)$, the number of primes less than x, and the complex zeros of $\zeta(z)$. Iqbal begins by describing a relationship between counting primes and powers of primes less than a given magnitude and uses weights. As an example, below is a table showing counts of primes and powers of primes, up to the number 40.

$x = 40$			
p	p^2	p^3	p^4
2,3,5,7,11,13,17,19, 23,29,31,37	*4,9,25*	*8,27*	*16*
$C(p)=12$	$C(p^2) = 3$	$C(p^3) = 2$	$C(p^4) = 1$

The last row shows the counts of the number, and if we count and weight the counts according to the powers we get the following:

$$\text{Weighted Count} = 12\left(\frac{1}{1}\right)+3\left(\frac{1}{2}\right)+2\left(\frac{1}{3}\right)+1\left(\frac{1}{4}\right)$$

$$= C(p)\left(\frac{1}{1}\right)+C\left(p^2\right)\left(\frac{1}{2}\right)+C\left(p^3\right)\left(\frac{1}{3}\right)+C\left(p^4\right)\left(\frac{1}{4}\right)$$

$$= \sum_{r=1}^{4}\frac{C\left(p^r\right)}{r}$$

The count function above $C(\ldots)$ is also a summation that counts the number of times the power occurs less than our value $x = 40$:

$$WC(x=40,r=4)=\sum_{r=1}^{4}\frac{C\left(p^{r}\right)}{r}=\sum_{r=1}^{4}\sum_{p^{r}\le 40}\frac{1}{r}$$

Iqbal generalizes the above line to the following,

(1) $$J(x)=\sum_{r=1}^{\infty}\sum_{p^{r}\le x}\frac{1}{r}$$

Let us recall the prime counting function $\pi(x)$:

(2) $$\pi(x)=\sum_{p\le x}1$$

We can substitute (2) into (1) and obtain the following

$$J(x)=\sum_{r=1}^{\infty}\sum_{p^{r}\le x}\frac{1}{r}=\sum_{r=1}^{\infty}\frac{1}{r}\sum_{p^{r}\le x}1$$

$$=\sum_{r=1}^{\infty}\frac{1}{r}\sum_{p\le x^{1/r}}1$$

$$=\sum_{r=1}^{\infty}\frac{1}{r}\pi\left(x^{1/r}\right)$$

(3) $$J(x)=\sum_{r=1}^{\infty}\frac{1}{r}\pi\left(x^{1/r}\right)$$

Observe above that (3) relates the prime counting function to the count of primes and powers of primes. Let us now try to solve for the prime counting function $\pi(x)$ in terms of the $J(x)$ function, to do this we will employ the Möbius function, $\mu(x)$, and its inversion. Think of the Möbius function as a function that "tracks" the number of prime factors of a number. If a number 'x' has repeated prime factors then the function will equal 0. If 'x' has distinct prime factors only then it will either equal +1 or -1:

$$\mu(x)=\begin{cases}0 & \text{if } x \text{ has repeated prime factors}\\ (-1)^{k} & \text{if } x \text{ has } k \text{ distinct prime factors}\end{cases}$$

Below is a table of example values of the Möbius function:

If we examine the table to the right; as an example, when x is 4 the Möbius function equals zero this is because the prime number repeats twice in the factorization of 4. When x is 6 the function equals +1 because 6 has 2 distinct non-repeating prime factors for is factorization; therefore, $k = 2$ in the function definition. Now if x is 13, well 13 is a prime number and it is the only prime number in its factorization; therefore, $k = 1$ in the function definition.

x	PRIME FACTORIZATION	$\mu(x)$	
1 =	1	$(-1)^0$ =	+1
2 =	2	$(-1)^1$ =	-1
3 =	3	$(-1)^1$ =	-1
4 =	2×2	0 =	0
5 =	5	$(-1)^1$ =	-1
6 =	2×3	$(-1)^2$ =	+1
7 =	7	$(-1)^1$ =	-1
8 =	2×2×2	0 =	0
9 =	3×3	0 =	0
10 =	2×5	$(-1)^2$ =	+1
11 =	11	$(-1)^1$ =	-1
12 =	2×2×3	0 =	0
13 =	13	$(-1)^1$ =	-1

Now how does the Möbius function help us solve for $\pi(x)$ function in (3)? Well notice that $J(x)$ and $\pi(x)$ are arithmetic functions, they are counting functions, for our purposes they are counting prime numbers. As it turns out we can apply the Möbius inversion relationship to arithmetic functions. Suppose you have two arithmetic function $g(x)$ and $f(x)$ and they are related in the following way below:

$$g(x) = \sum_{d|x} f(d) \quad \text{for } x \in \mathbb{Z}_{\geq 1}$$

With Möbius inversion we can swap the $g()$ and $f()$ functions with a simple modification to the original summation term:

$$f(x) = \sum_{d|x} \mu(d) g(n/d) \quad \text{for } x \in \mathbb{Z}_{\geq 1}$$

If we use the inversion transformation described above with (3) we get,

(4) $$\pi(x) = \sum_{r=1}^{\infty} \frac{\mu(r)}{r} J\left(x^{1/r}\right)$$

Iqbal from here establishes a link between $\pi()$, $J()$, and the ζ-function. Let us recall the Euler Definition of the ζ-function:

(5) $$\zeta(s) = \prod_p \frac{1}{\left(1 - \frac{1}{p^s}\right)} \quad \text{where } s \in \mathbb{C}$$

Then we take the log() of (5) to get,

$$\log \zeta(s) = \sum_p \log\left(\frac{1}{\left(1-\frac{1}{p^s}\right)}\right)$$

$$= -\sum_p \log\left(1-\frac{1}{p^s}\right)$$

(6) $$= -\sum_p \log\left(1-p^{-s}\right)$$

Now Iqbal does something interesting, he turns the log() in (6) into an infinite series, recall the Taylor Series of the log() function:

$$\log(x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}(x-1)^k$$

Let $x = 1-p^{-s}$ in the above Taylor Series we obtain,

$$\log(1-p^{-s}) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}\left(1-p^{-s}-1\right)^k$$

$$= \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}(-1)^k p^{-ks}$$

(7) $$= \sum_{k=1}^{\infty} \frac{(-1)^{2k+1}}{k} p^{-ks}$$

Substitute (7) into (6) and we get,

$$\log \zeta(s) = -\sum_p \log\left(1-p^{-s}\right)$$

$$= -\sum_p \sum_{k=1}^{\infty} \frac{(-1)^{2k+1}}{k} p^{-ks} = \sum_p \sum_{k=1}^{\infty} \frac{(-1)^{2k+2}}{k} p^{-ks}$$

(8) $$= \sum_p \sum_{k=1}^{\infty} \frac{1}{k} p^{-ks}$$

Let us now consider the following integral,

$$\int_a^{\infty} x^{-s-1}dx = \frac{x^{-s}}{-s}\Big|_a^{\infty}$$
$$= \frac{\infty^{-s}}{-s} + \frac{a^{-s}}{s} = 0 + \frac{a^{-s}}{s}$$

Hence,

$$\int_a^{\infty} x^{-s-1}dx = \frac{a^{-s}}{s} \quad \Rightarrow \quad a^{-s} = s\int_a^{\infty} x^{-s-1}dx$$

In the integral above let $a = p^k$ to get,

(9) $$p^{-ks} = s\int_{p^k}^{\infty} x^{-s-1}dx$$

Substitute (9) into (8) to and you obtain,

$$\log\zeta(s) = s\sum_p\sum_{k=1}^{\infty}\frac{1}{k}\int_{p^k}^{\infty} x^{-s-1}dx$$

Therefore,

$$\frac{\log\zeta(s)}{s} = \sum_p\sum_{k=1}^{\infty}\frac{1}{k}\int_{p^k}^{\infty} x^{-s-1}dx$$

(10) $$= \sum_{k=1}^{\infty}\frac{1}{k}\sum_p\int_{p^k}^{\infty} x^{-s-1}dx$$

Let us examine the first three iterations of (10), below is a table,

$k = 1$	$\frac{1}{1}\left(\int_2^{\infty} x^{-s-1}dx + \int_3^{\infty} x^{-s-1}dx + \int_5^{\infty} x^{-s-1}dx\cdots\right)$
$k = 2$	$\frac{1}{2}\left(\int_{2^2}^{\infty} x^{-s-1}dx + \int_{3^2}^{\infty} x^{-s-1}dx + \int_{5^2}^{\infty} x^{-s-1}dx\cdots\right)$
$k = 3$	$\frac{1}{3}\left(\int_{2^3}^{\infty} x^{-s-1}dx + \int_{3^3}^{\infty} x^{-s-1}dx + \int_{5^3}^{\infty} x^{-s-1}dx\cdots\right)$

Notice that we are taking a weighted summation of the integrals and the summation is over powers of prime numbers. Essentially we have our weighted powers of prime numbers counting function $J(x)$. This means we can express our expression (10) in the following way,

(11) $$\frac{\log \zeta(s)}{s} = \int_0^\infty J(x) x^{-s-1} dx$$

Now, Iqbal uses the Mellin Transform to solve for our $J(x)$ function in (11). Below is the definition of the transform.

$$\varphi(s) = \int_0^\infty x^{s-1} f(x) dx \quad \Leftrightarrow \quad f(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} x^{-s} \varphi(s) ds$$

If we let $\varphi(s) = \frac{\log \zeta(s)}{s}$ and $f(x) = J(x)$ we get,

(12) $$J(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\log \zeta(s)}{s} x^{-s} ds$$

Next, let us consider Hadamard's infinite product representation of the ζ-function,

$$\zeta(s) = \frac{\pi^{s/2}}{2(s-1)\Gamma(1+s/2)} \prod_w \left(1 - \frac{s}{w}\right) \quad \text{where } \zeta(w) = 0 \text{ and } \operatorname{Re}(w) = \frac{1}{2}$$

This representation expresses the ζ-function in terms of its non-trivial zeros w. If we combine Hadamard's representation with (12) we obtain the following:

$$J(x) = li(x) - \sum_w li\left(x^w\right) - \log(2) + \int_x^\infty \frac{dt}{t\left(t^2-1\right)\log(t)}$$

Recall (4), $\pi(x)$ is determined from $J(x)$ and the line above shows that the counting function $J(x)$ is based upon the non-trivial zeros of the ζ-function. In essence, Iqbal has shown that the prime-counting function depends on the non-trivial zeros of the ζ-function.

2. The Number of Roots of the ζ-Function Between 0 and T

In Riemann's paper "On the Number of Primes Less Than a Given Magnitude," Riemann makes a curious statement. He says the number of non-trivial roots of $\zeta(s)$ between 0 and T is,

(13) $$= \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi}$$

Let us take (13) and work backwards,

$$= \frac{T}{2\pi}\left(\log\frac{T}{2\pi} - 1\right) = \frac{T}{2\pi}\left(\log\frac{T}{2\pi} - \log 0\right)$$

$$\#\text{ of Zeros } = \frac{T}{2\pi}\int_0^{T/2\pi}\frac{dt}{t}$$

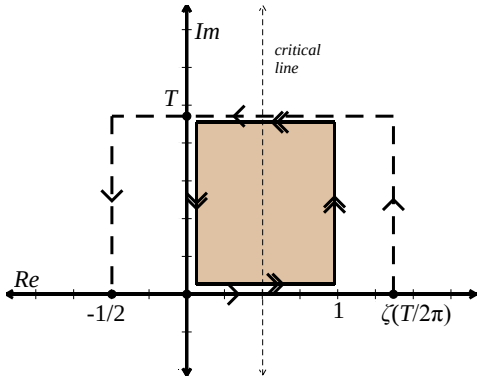
A more precise representation would be:

(14) $$\#\text{ of Zeros} = \sum_{\substack{\zeta(w)=0 \\ \mathrm{Im}(w)\in[0,T]}} 1 \sim \frac{T}{2\pi}\int_0^{T/2\pi}\frac{dt}{t}$$

The summation above counts the number of zeros, w, of the ζ-function on the interval $0 \le \mathrm{Im}(w) \le T$. Edwards' explanation to how Riemann knew this fact, was a result of Riemann being a "master" of evaluating integrals. Riemann's argument was that the number of roots in a closed region was equal to an integral over a closed region. In (14) if we let $t = \zeta(s)$ we get,

(15) $$\#\text{ of Zeros} = \frac{T}{2\pi}\int_{-1/2}^{\zeta(T/2\pi)}\frac{\zeta'(s)ds}{\zeta(s)}$$

But this integral is over a closed rectangular region shown below.

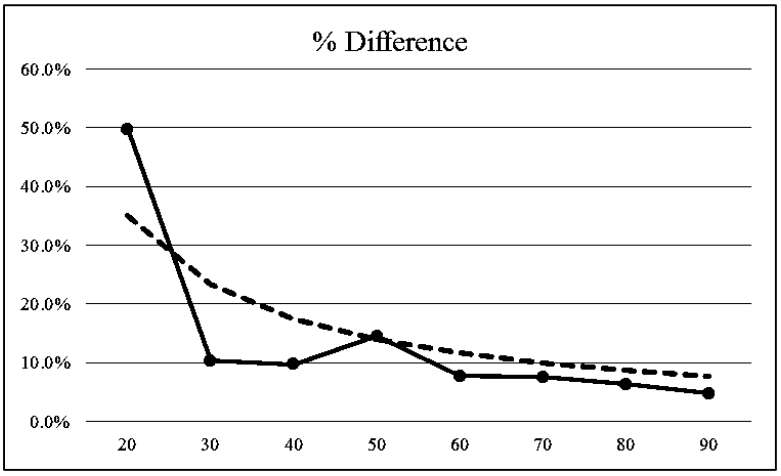


The integral in (15) corresponds to the dashed contour in the graph above, but the number of zeros will still be the same if we choose a smaller contour to integrate over and closer to the critical line where Riemann hypothesized the non-trivial zeros would occur. If we take this into consideration we get,

$$\#\text{ of Zeros} = \left|\frac{T}{2\pi i}\oint_C\frac{\zeta'(s)ds}{\zeta(s)}\right|$$

Where, $C:\{0 \le \mathrm{Re}(s) \le 1, 0 \le \mathrm{Im}(s) \le T\}$. Using Mathematica we can verify this approximation of the number of non-trivial zeros less than *T*. Below is a table that compares the approximation that Riemann predicted versus the actual count that Mathematica determines.

T	$\frac{T}{2\pi}\log\frac{T}{2\pi}-\frac{T}{2\pi}$	**Mathematica Zero Count**	**% Difference.**
20	0.5024	1	49.75%
30	2.6896	3	10.34%
40	5.4176	6	9.71%
50	8.5477	10	14.52%
60	11.9984	13	7.70%
70	15.7155	17	7.56%
80	19.6607	21	6.38%
90	23.8054	25	4.78%



The graph above shows that the percent difference decreases by $\sim T^{-1}$; the solid line graph corresponds to the values in the table of % differences above, and the dashed line corresponds to the function 6/*T*. Therefore, for large values of *T,* Riemann's approximation is fairly accurate.

3. An Alternate Form of the Riemann Hypothesis Using the Error.

Let us explore an alternate form of the Riemann Hypothesis as presented by Edwards [Edwards, p. 19]. The hypothesis is equivalent to saying: given an error $\epsilon > 0$, where the error is from the Prime Counting function being approximated by the logarithmic integral *Li*(*x*), is less than $x^{-\frac{1}{2}+\epsilon}$. In other words if,

(16) $$\pi(x) \sim Li(x)$$

Then,

(17) $$\left|\pi(x) - Li(x)\right| \le C_{\epsilon} x^{\frac{1}{2}+\epsilon}$$

Recall,

$$Li(x) = \int_2^x \frac{dt}{\log(t)}$$

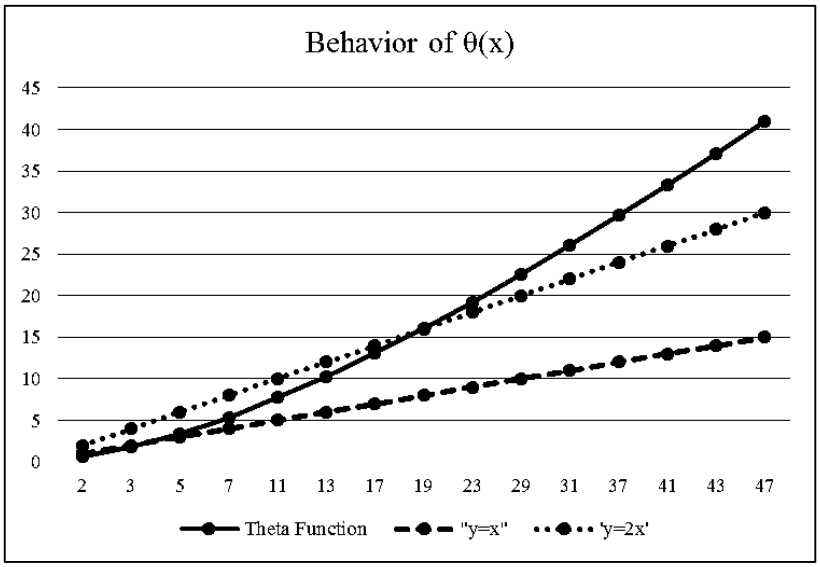
The logarithmic integral can also be approximated for sufficiently large *x*,

$$Li(x) \sim \frac{x}{\log(x)}$$

From Edwards [p. 91], we can use the θ-function. Recall that the θ-function is the summation of the logarithms of primes, *p*, less than a value of '*x*.'

$$\theta(x) = \sum_{p<x} \log(p) \sim Cx$$

To the right is a graph of the θ-function, the curved plot is the θ-function, the other two straight line plots are of the linear function $y = x$ and $y = 2x$. For large values of *x* the θ-function behaves like a *const*. times *x*.



If we now consider the error in the θ-function and *x* we obtain,

$$\theta(x) - x = \int_a^x \log(t) d[\pi(t) - Li(t)] + \text{Constant}$$

For the line above we can use integration by parts. Let $u = \log(t)$ and $dv = d[\pi(t) - Li(t)]$, then $du = 1/t$ and $v = \pi(t) - Li(t)$:

$$\int_a^x \log(t) d[\pi(t) - Li(t)] + Const. = uv - \int v\, du + Const.$$

$$= \log(t)[\pi(t) - Li(t)]\Big|_a^x - \int_a^x \frac{\pi(t) - Li(t)}{t} dt + Const.$$

(18)
$$\theta(x) - x = \log(x)[\pi(x) - Li(x)] - \int_a^x \frac{\pi(t) - Li(t)}{t} dt + Const.$$

Now let us take the magnitude of (18) to get,

$$\begin{aligned}
\left|\theta(x)-x\right| &= \left|\log(x)[\pi(x)-Li(x)]-\int_a^x \frac{\pi(t)-Li(t)}{t}dt+Const.\right| \\
&\le \left|\log(x)[\pi(x)-Li(x)]\right|+\left|\int_a^x \frac{\pi(t)-Li(t)}{t}dt\right|+Const. \\
&\le \log(x)\left|\pi(x)-Li(x)\right|+\int_a^x \frac{\left|\pi(t)-Li(t)\right|}{t}dt+Const. \\
&= \log(x)x^{1/2+\epsilon}+\int_a^x \frac{t^{1/2+\epsilon}}{t}dt+Const. \\
&= \log(x)x^{1/2+\epsilon}+\int_a^x t^{-1/2+\epsilon}dt+Const. \\
&\le \log(x)x^{1/2+\epsilon}+x^{1/2+\epsilon}+Const.
\end{aligned}$$

Therefore,

(19) $$\left|\theta(x)-x\right| \le \log(x)x^{1/2+\epsilon}+x^{1/2+\epsilon}+Const. \le x^{1/2+2\epsilon}$$

Now let us consider Chebyshev's formula,

$$\psi(x)=\sum_{k=1}^{\infty}\theta\left(x^{1/k}\right)=\theta\left(x\right)+\theta\left(x^{1/2}\right)+\theta\left(x^{1/3}\right)+...$$

This immediately implies that,

(20) $$\theta(x)<\psi(x)<\theta(x)+\theta\left(x^{1/2}\right)\frac{\log(x)}{\log(2)} \sim \theta(x)+x^{1/2}\log(x)$$

The above inequality can be establish because Chebyshev's summation is over θ-functions, so numerically it has to be greater than the θ-function. Secondly, in Chebyshev's function there are terms that extend to powers of ½, 1/3, ¼…and so on, these essentially go to 0 for large *x*. This means that the Chebyshev function is less than the first term of its summation plus the maximum factor of the θ-function for the power ½. The far right term of the inequality is asymptotic to one factor of the θ-function plus a logarithmic factor, for extremely large *x*, the log(2) term is insignificant compared to the logarithm of infinity. Now if we consider the following,

$$\psi(x)-x \sim \theta(x)-x+x^{1/2}\log(x)$$

Let us take the magnitude of the above line to get,

(21) $$|\psi(x)-x| < |\theta(x)-x| + x^{1/2+\epsilon}\log(x)$$

Combine this with (19) and we obtain,

(22) $$|\psi(x)-x| < x^{1/2+\epsilon} + x^{1/2+\epsilon}\log(x) < x^{1/2+2\epsilon}$$

This is true for extremely large values of *'x'*. Let us now consider the logarithm of the ζ-function and then differentiate to get,

$$\log\zeta(s) = \int_0^\infty x^{-s}dJ(x)$$

$$\frac{d}{ds}\log\zeta(s) = \frac{d}{ds}\int_0^\infty x^{-s}dJ(x)$$

$$\frac{\zeta'(s)}{\zeta(s)} = -\int_0^\infty x^{-s}\underbrace{\log x\, dJ(x)}_{d\psi(x)}$$

$$-\frac{\zeta'(s)}{\zeta(s)} = \int_0^\infty x^{-s}d\psi(x)$$

Again we will use integration by parts for the integral on the right side of the equality for the above line. Let $u = x^{-s}$, $du = -sx^{-s-1}dx$, $dv = d\psi(x)$, $v = \psi(x)$ to get,

$$-\frac{\zeta'(s)}{\zeta(s)} = \int_0^\infty x^{-s}d\psi(x) = uv - \int v\,du$$

$$= x^{-s}\psi(x)\Big|_0^\infty - \int_0^\infty -sx^{-s-1}\psi(x)\,dx$$

(23) $$-\frac{\zeta'(s)}{\zeta(s)} = s\int_1^\infty x^{-s-1}\psi(x)\,dx$$

The integral bounds were changed from 0 to 1 because at 0 the function is undefined. Next let us consider the quantity 1/(1 – s) and work backwards,

$$-\frac{1}{1-s} = \left(\frac{1}{1-s}\right)\left(\frac{1}{\infty^s}\right) - \left(\frac{1}{1-s}\right)\left(\frac{1}{1^s}\right)$$

$$= \frac{\infty^{-s}}{1-s} - \frac{1^{-s}}{1-s} = \frac{x^{-s}}{1-s}\Big|_1^\infty$$

(24) $$-\frac{1}{1-s}=\int_1^\infty x^{-s}dx=-1+s\int_1^\infty x^{-s-1}x\,dx$$

Now take (23) and (24) and subtract the left hand sides and subtract the right hand sides together to get,

$$-\left(\frac{\zeta'(s)}{\zeta(s)}-\frac{1}{1-s}\right)=s\int_1^\infty x^{-s-1}\psi(x)\,dx-\left(-1+s\int_1^\infty x^{-s-1}x\,dx\right)$$

$$-\left(\frac{\zeta'(s)}{\zeta(s)}+\frac{1}{s-1}\right)=1+s\int_1^\infty x^{-s-1}\left[\psi(x)-x\right]dx$$

$$-\frac{d}{ds}\log\left[\zeta(s)(s-1)\right]=1+s\int_1^\infty x^{-s-1}\left[\psi(x)-x\right]dx$$

Recall from (22) that $\left|\psi(x)-x\right|<x^{1/2+2\epsilon}$; therefore, the integral on the right hand side will converge as we let x get arbitrarily large and $\epsilon\to 0$. What this means is that we have established a definition of the ζ-function, the right hand side, using Chebyshev's function for primes. And since $\psi(x)\sim x$ and its relative error $\epsilon\to 0$ as $x\to\infty$, then the relative error value in the prime counting rule and the logarithmic integral is less than $x^{1/2+\epsilon}$.

4. An Elementary Problem Equivalent to the Riemann Hypothesis by Lagarias.

Lagarias' paper begins by presenting a problem and then he proves that the problem is equivalent to the Riemann Hypothesis. He considers the Harmonic numbers $H_n=\sum_{j=1}^{n}1/j$, and states that,

(25) $$\sigma(n)=\sum_{d|n}d\le H_n+\exp(H_n)\log(H_n)$$

The σ-function on the right is the summation of the divisors of 'd'; for instance, $\sigma(8)=1+2+4+8=15$. Now Lagarias makes use of proof by contradiction and states that if the Riemann Hypothesis is false then the following inequality must hold,

(26) $$\sum_{d|n}d\ge e^{\gamma}n\log\log n+\frac{Cn\log\log n}{(\log n)^{\beta}}$$

Where $0<\beta<½$ and $C>0$. Recall the definition of the Euler-Mascheroni constant, $\gamma=\lim_{n\to\infty}\left(H_n-\log n\right)=0.57721...$

Lagarias begins by proving the following,

(27) $$\exp(H_n)\log(H_n) \geq e^{\gamma} n \log\log n$$

We will utilize the integer function $\lfloor t \rfloor = \sum_{1 \leq r \leq t} 1$ and $t = \lfloor t \rfloor + \{t\} = Integer + Fractional$, let us consider the integral,

$$\int_1^n \frac{\lfloor t \rfloor}{t^2} dt = \int_1^n \frac{1}{t^2} \left(\sum_{1 \leq r \leq t} 1 \right) dt$$

For the statement above we can move the summation outside the integral and change the integral bounds and summation bounds. The value of *t* in the summation bound is varying from 1 to *n* in the integral. If we move the summation outside the integral, the lower bound of the integral will depend on the value of *r* that is varying. We get,

$$\begin{aligned} \int_1^n \frac{\lfloor t \rfloor}{t^2} dt &= \sum_{1 \leq r \leq n} \int_r^n \frac{dt}{t^2} \\ &= \sum_{r=1}^{n} \left(\frac{1}{r} - \frac{1}{n} \right) = \sum_{r=1}^{n} \frac{1}{r} - \sum_{r=1}^{n} \frac{1}{n} = \\ &= H_n - n\left(\frac{1}{n} \right) \end{aligned}$$

(28) $$\int_1^n \frac{\lfloor t \rfloor}{t^2} dt = H_n - 1$$

Solve for the harmonic number in (28) gets,

$$H_n = 1 + \int_1^n \frac{\lfloor t \rfloor}{t^2} dt$$

Remember that we defined $t = \lfloor t \rfloor + \{t\}$; therefore, substitute $\lfloor t \rfloor = t - \{t\}$ into the line above to obtain,

$$\begin{aligned} H_n &= 1 + \int_1^n \frac{t - \{t\}}{t^2} dt = 1 + \int_1^n \frac{dt}{t} - \int_1^n \frac{\{t\} dt}{t^2} \\ &= 1 + \log n - \int_1^n \frac{\{t\} dt}{t^2} \end{aligned}$$

(29) $$H_n = \log n + 1 - \int_1^n \frac{\{t\}\,dt}{t^2}$$

Below is some manipulation of (29),

$$\begin{aligned}
H_n &= \log n + 1 - \int_1^n \frac{\{t\}\,dt}{t^2} - \int_n^\infty \frac{\{t\}\,dt}{t^2} + \int_n^\infty \frac{\{t\}\,dt}{t^2} \\
&= \log n + 1 - \left(\int_1^n \frac{\{t\}\,dt}{t^2} + \int_n^\infty \frac{\{t\}\,dt}{t^2} \right) + \int_n^\infty \frac{\{t\}\,dt}{t^2} \\
&= \log n + 1 - \int_1^\infty \frac{\{t\}\,dt}{t^2} + \int_n^\infty \frac{\{t\}\,dt}{t^2} \\
&= \log n + \left(\frac{1}{1} - \frac{1}{\infty} \right) - \int_1^\infty \frac{\{t\}\,dt}{t^2} + \int_n^\infty \frac{\{t\}\,dt}{t^2} \\
&= \log n + \int_\infty^1 \frac{-dt}{t^2} - \int_1^\infty \frac{\{t\}\,dt}{t^2} + \int_n^\infty \frac{\{t\}\,dt}{t^2} \\
&= \log n + \int_1^\infty \frac{dt}{t^2} - \int_1^\infty \frac{\{t\}\,dt}{t^2} + \int_n^\infty \frac{\{t\}\,dt}{t^2} \\
&= \log n + \int_1^\infty \left(\frac{1 - \{t\}}{t^2} \right) dt + \int_n^\infty \frac{\{t\}\,dt}{t^2} \\
&= \log n + \int_1^\infty \left(\frac{1 - t + \lfloor t \rfloor}{t^2} \right) dt + \int_n^\infty \frac{\{t\}\,dt}{t^2} \\
&= \log n + \int_1^\infty \left(\frac{1 - t + \lfloor t \rfloor}{t^2} \right) dt + \int_n^\infty \frac{\{t\}\,dt}{t^2} \\
&= \log n + \int_1^\infty \left(\frac{1 + \lfloor t \rfloor}{t^2} - \frac{1}{t} \right) dt + \int_n^\infty \frac{\{t\}\,dt}{t^2}
\end{aligned}$$

In the last line above, $\int_1^\infty \left(\frac{1 + \lfloor t \rfloor}{t^2} - \frac{1}{t} \right) dt$ converges to γ; therefore,

(30) $$H_n = \log n + \gamma + \int_n^\infty \frac{\{t\}\,dt}{t^2}$$

From (30) we can conclude that,

(31) $$H_n > \log n + \gamma$$

Now exponentiate both sides of the inequality (31) to get,

$$\begin{aligned}\exp\left(H_n\right) &> e^{\log n+\gamma} \\ &= e^{\log n}e^{\gamma} \\ &= ne^{\gamma}\end{aligned}$$

(32) $$\exp\left(H_n\right) > ne^{\gamma}$$

Also from (31),

$$H_n > \log n$$

(33) $$\log H_n > \log\log n$$

Now multiply (32) and (33) together to get (27). Next, Lagarias considers the definition of the Euler-Mascheroni constant,

$$\gamma = \lim_{n\to\infty}\left(H_n - \log n\right) = \lim_{n\to\infty}\left(H_n - \log(n+1)\right)$$

Notice above that we have also included $n+1$, this is because, the limit will still converge to the constant because $\infty + 1 = \infty$. Therefore, we can conclude the following,

$$\gamma \geq H_n - \log(n+1)$$

We exponentiate both sides of the inequality above to get,

$$\begin{aligned}\exp(\gamma) &\geq e^{H_n-\log(n+1)} \\ &= e^{H_n}e^{-\log(n+1)} \\ &= \exp\left(H_n\right)\frac{1}{n+1} \\ \exp(\gamma) &\geq \exp\left(H_n\right)\frac{1}{n+1}\end{aligned}$$

(34) $$\exp\left(H_n\right) \leq e^{\gamma}(n+1)$$

From (29) we can write the following inequality,

(35) $$H_n < \log n + 1 < \frac{n}{\log n}$$

Take the logarithm on both sides of (35) to get,

(36) $$\log\left(H_n\right) < \log\left(\log n + 1\right)$$

Using some manipulation, we can rewrite (36),

$$\begin{aligned}\log\left(H_n\right) &< \log\left(\log n+1\right)\\ &= \log\left(\log n+\frac{\log n}{\log n}\right)=\log\left(\log n\left(1+\frac{1}{\log n}\right)\right)\\ &= \log\log n+\log\left(1+\frac{1}{\log n}\right)\\ &< \log\log n+\frac{1}{\log n}\end{aligned}$$

(37) $$\log\left(H_n\right) < \log\log n+\frac{1}{\log n}$$

Now, let us multiply (34) and (37) to obtain,

$$\begin{aligned}\exp\left(H_n\right)\log\left(H_n\right) &< e^{\gamma}(n+1)\left(\log\log n+\frac{1}{\log n}\right)\\ &= e^{\gamma}n\log\log n+e^{\gamma}\log\log n+\frac{e^{\gamma}n}{\log n}+\frac{e^{\gamma}}{\log n}\\ &= e^{\gamma}n\log\log n+\frac{e^{\gamma}n}{\log n}+e^{\gamma}\left(\log\log n+\frac{1}{\log n}\right)\end{aligned}$$

For the line above we can combine the following inequality with the parenthetical expression above.

$$\log\log n+\frac{1}{\log n}\le\frac{n}{2\log n}$$

To get,

$$\begin{aligned}\exp\left(H_n\right)\log\left(H_n\right) &< e^{\gamma}n\log\log n+\frac{e^{\gamma}n}{\log n}+\frac{e^{\gamma}n}{2\log n}\\ &= e^{\gamma}n\log\log n+e^{\gamma}n\left(\frac{1}{\log n}+\frac{1}{2\log n}\right)\\ &= e^{\gamma}n\log\log n+\frac{3e^{\gamma}n}{2\log n}\end{aligned}$$

(38) $$\exp\left(H_n\right)\log\left(H_n\right) < e^{\gamma}n\log\log n+\frac{3e^{\gamma}n}{2\log n}$$

We add (35) and (38) together to get,

$$H_n + \exp\left(H_n\right)\log\left(H_n\right) < e^{\gamma} n \log\log n + \frac{3e^{\gamma} n}{2\log n} + \frac{n}{\log n}$$

$$= e^{\gamma} n \log\log n + \frac{n}{\log n}\left(\frac{3e^{\gamma}}{2} + 1\right)$$

$$< e^{\gamma} n \log\log n + \frac{4n}{\log n}$$

(39) $$H_n + \exp\left(H_n\right)\log\left(H_n\right) < e^{\gamma} n \log\log n + \frac{4n}{\log n}$$

Lagarias utilizes Gronwall's Theorem which relates σ(n) to the Euler-Mascheroni constant,

$$\lim_{n\to\infty} \frac{\sigma(n)}{n\log\log n} = e^{y} \Rightarrow \frac{\sigma(n)}{n\log\log n} < e^{y}$$

$$\sigma(n) < e^{y} n \log\log n$$

(40) $$\sum_{d|n} d < e^{y} n \log\log n$$

If we combine (39) with (40) we get,

(41) $$\sum_{d|n} d < H_n + \exp\left(H_n\right)\log\left(H_n\right) < e^{y} n \log\log n$$

But (41) contradicts (26) and (27) because in (26) and (27) the inequality is in the wrong direction. Statement (41) verifies the original problem stated by Lagarias; therefore, the original problem is an equivalent statement to the Riemann Hypothesis.

REFERENCES

[1] Edwards, H. M. (**1974**) *Riemann's Zeta Function*, Academic Press (New York).

[2] Lagarias, Jeffrey C. (**2002**) *An Elementary Problem Equivalent to the Riemann Hypothesis*, Mathematical Association of America (U.S.A.).

[3] Iqbal, Amer. (**2005)** *Prime Numbers and the Riemann Zeta Function*, University of Seattle Department of Mathematics (Seattle, WA).